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3D RECONSTRUCTION PIPELINE FOR NEURAL CIRCUITRY

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Abstract. The increasing resolution capabilities in electron microscopy have propelled a new frontier in neurobiology and computational science: imaging and segmenting the brain at the nano level to map its intricate neuronal connections. However, a single cubic millimeter of brain tissue in a mouse requires around 2000 terabytes of imaging data, posing significant computational challenges. In this study, we introduce a novel three-step approach for 3D instance segmentation in connectomics, leveraging a U-Net-based architecture enhanced with new data augmentation techniques and a CPU-parallelized processing pipeline. Our model integrates smooth deformations, overlap displacement, and skip connections to improve accuracy while reducing dependency on labor-intensive manual annotations. We compare our network's performance with U-Net and DeepLab networks across diverse datasets, including multiple neural systems and species, achieving superior segmentation accuracy and generalizability. Results demonstrate a marked improvement in cell membrane classification and overall segmentation robustness, underscoring our approach's potential to streamline and scale the connectomics pipeline, paving the way towards a comprehensive blueprint of the brain's connectivity.

Keywords: Deep Learning, Connectome, Segmentation, U-net, CNN-LSTM, Electron Microscope Image.

1. Introduction

With the recent progress in electron microscopy, neurobiologists and computer scientists urgently to image the brain circuitry at the nano level to get a formal and regular blueprint. However, a single cubic millimeter of brain tissue in a mouse cortical column, including around millions of neurons and billions of synapses, requires 2000 terabytes of image volume! For the human cortical column, it will become an enormous number. Aiming to solve this 3D instance segmentation problem and try to distinguish neuron morphology as much as we could, Connectomics has been founded.

The term connectome was first used independently by Olaf Sporns and Patrick Hagmann to delineate "a comprehensive structural description of the network of elements and connections forming the human brain" [1,2]. Ever since, the connectome is used to refer to any systematic account of connections, from local circuits to entire nervous systems [2]. Although the term was first adapted in 2005, the idea of carefully studying the anatomy of neural circuits alongside their function can be traced back to the godfather of modern neuroscience, Santiago Ramon y Cajal. Structure and function were finely intertwined in Ramon y Cajal's studies, who predicted fundamental functional properties of the nervous system based on detailed observations of its anatomical structure, which he documented in extraordinary drawings [2].

It is however perhaps one of the hardest 3D instance segmentation problems and is not well solved yet. The current methods cannot satisfy both scalable and accurate [3,4,5]. The drawback of machine learning is that the geometry and branching topology of neurons in the brain is hugely complex. In fact, so far, all the connectomic field study depends on massive human effort, usually hundreds of people per project, which are annotators [6,7] to do the annotating, proofreading, and correcting segment of every single neuron. A direct example is the current human brain connectome group, they have 87 annotators with 4,000 human working hours, and just for correcting the error generated by the automatic segmentation [8]. However, compared to 20,000 pure hands annotation hours, without using any machine help in the early study of the mammalian retina, which was a great improvement already [8]. Still, 4,000 human hours investment in just a single reconstruction volume makes our goal (building the human brain blueprint) looks impossible. It's thus important to bring a more effective way to cut the number down as much as we can.

In this project, we adapted a new approach based on the U-Net convolution networks [18], to facilitate connectomic reconstruction. It's different from other reconstruction methods, which need human sloppy annotation into inputs of their pipeline, this method will rely on computation resources to generate semi-auto/auto segmentation results as different needs. Moreover, in order to solve the accuracy problem, this study proposed two more data augment methods, smooth deformations, and overlap displacement besides U-net unique features. My network architecture also integrates a new mechanism, skip connection, to help target area classification to improve the accuracy performance. To increase the speed of the whole process, this study here for the first time adapts the

CPU parallel computation method [19], to maximize performance on multicore CPU systems.

For evaluation, we conducted experiments to compare my network with two widely used convolution networks in the connectomics field, U-net and Deep Lab net. The experiment results show that my network significantly outperforms the given datasets. Moreover, to gain further insight into the successes and limitations of this study, we adapted our data augment methods to U-net and Deep Lab net [18], the result shows similar problems existed in different networks, and all three nets significantly improved after adaptation. Finally, the current study shows high generalizability across datasets, we were not only used mouse brain dataset, but also the mouse muscle (same specie but different neural system), human brain, octopus' brain, and c'elegans' brain (different species but same neural system). The main contributions of this project are summarized below. we provide 1. a new and effective network for reconstructing biomedical imaging. A significant improvement compared to the current popular method: U-net and DeepLab net. 2. novel data augment methods, smooth deformations, and overlap displacement to reduce the human hours in preparing ground truth and increase the accuracy. 3. a high generalizability model across datasets. Proved by using the same specie but different neural systems and different species but the same neural system. Section 7.2.

2. Literature Review

Currently, more and more research groups have suggested end-to-end algorithms. Januszewski et al. proposed flood-filling networks with "a recurrent procedure for iterating the network inference dynamics over multiple overlapping fields of view in order to segment arbitrarily large objects that are initialized from a single seed pixel". The core idea here is that a single object is masked and a fully convolutional network discriminates whether a pixel pertains to such an object or not, and then this procedure is repeated many times in different directions to saturate the volume. This end-to-end approach was pursued independently also by Meirovitch et al. with MaskExtend. Although these algorithms improved many benchmark accuracies, they are limited by the fact that they track one object at a time, thus appearing impractical for large-scale Connectomics. One of the first steps towards a large-scale Connectomics approach was taken by Cross-classification clustering, by Meirovitch et al., where multiple neurites are tracked at the same time by transferring segmented objects from one image to the next with a computationally inexpensive method.

Opting for a fully automatic approach, i.e. excluding human intervention from the pipeline, usually has as one of its prerequisites a well- (if not perfectly) aligned stack of good quality images, which is not always a feasible goal. Furthermore, these complex models need to be tailor-trained and fine-tuned on each dataset, posing a new challenge to life science researchers who might not have substantial computer science expertise.

For these reasons, numerous packages have been proposed allowing the single researcher to handle the problem of segmentation autonomously. Most of them rely on a semi-automatic solution, hence the inclusion of human expertise in the process. Of particular relevance is Ilastik , a successful tool which relies on shallow machine learning techniques (in particular random forest) to perform swift interactive and reactive pixel-classification tasks. The use of machine learning algorithms, as opposed to deep learning, represents a compromise between training time and final accuracy (both less than those associated with deep learning methods).

Examples of solutions leveraging on a deep learning framework have been suggested by the Helmstaedter group with SegEM, by the Pfister group with ICON, which uses sparse labeling and proposes one of the first interactive deep learning frameworks for Connectomics, and by UNI-EM, which provides a complete framework for the whole connectomic reconstruction pipeline. However, speed is the priority factor among those methods, which makes the accuracy dramatically different. To balance the outcome, a classical three steps approach has been considered nowadays (i.e., see U-net and DeepLab net), which is also the main topic of this study. The approach is to first transform each image into a so-called boundary map, where each pixel is classified either as a boundary or as a non-boundary. This gives rise to a semantic segmentation of the input image (i.e., each pixel is classified in one class out of a predefined number of classes). While a simple approach consists in thresholding images, a more recent approach adopts supervised learning algorithms (such as convolutional neural networks) for boundary detection, which yields to better accuracy. In a second step, the semantic segmentation is transformed into an instance segmentation (i.e., each object has its own ID) by a flood-filling algorithm, for example, Watershed. Because many of these algorithms yield to an over-segmentation of the image, a third step is often required, which agglomerates over-segmented fragments in the final objects.

3. Dataset Description

The dataset used in this study consists of high-resolution electron microscopy (EM) images acquired through a well-documented and reproducible pipeline. These images are sourced from the AC3 and AC4 EM and segmentation stacks provided by the Kasthuri 2015 Cell paper, which are openly available via the VAST Lite Viewer platform(https://lichtman.rc.fas.harvard.edu/vast). The dataset includes samples from diverse biological systems such as:

- Mouse brain cortex and interscutularis muscle,

- Octopus brain,
- *C. elegans* brain,
- Human brain cortex.

The dataset plays a crucial role in connectomics research, providing nanoscale resolution imaging that enables detailed reconstruction of neural architectures. The following steps outline the process of making the data practically usable (Figure 1):

1. Sample Preparation Biological tissues from various sources are prepared using perfusion techniques to fix and preserve structural integrity. These tissues are first examined under **light microscopy** to ensure proper visualization and alignment for subsequent processing.

2. Tissue Processing and Imaging

- Tissues are processed using **micro-CT imaging** to confirm spatial integrity and guide sectioning into ultra-thin slices.
- The sections are converted into wafers using **wafer-making** techniques, ensuring uniform thickness and spatial consistency.

3. Electron Microscopy and Data Alignment

- Each wafer is imaged at nanoscale resolution using **EM imaging**, capturing detailed structural features.
- Images are then aligned and stitched to reconstruct continuous 3D representations of the neural structures.

4. 3D Reconstruction and Data Usability

- The aligned image stacks are reconstructed into 3D volumes for connectomics research.
- Preprocessing steps such as contrast normalization, noise reduction, and segmentation are applied to ensure high usability in downstream analysis.

Additionally, the data collection includes batch sampling methods with appropriate collation to maintain uniform input dimensions for computational modeling. These methods ensure biological relevance while optimizing the dataset for training machine learning algorithms.

This pipeline integrates cutting-edge technologies across biological imaging, data processing, and computational reconstruction, making the dataset indispensable for addressing key connectomics research questions.

4. Evaluation Metric

In this study, we focus on the segmentation and reconstruction of cell membranes from electron microscopy (EM) images, aiming to generate high-quality neuronal connectivity maps. To comprehensively quantify and evaluate the performance of our proposed pipeline—from membrane probability map generation, over-segmentation aggregation, error detection and reconstruction to final neuron-level segmentation—we

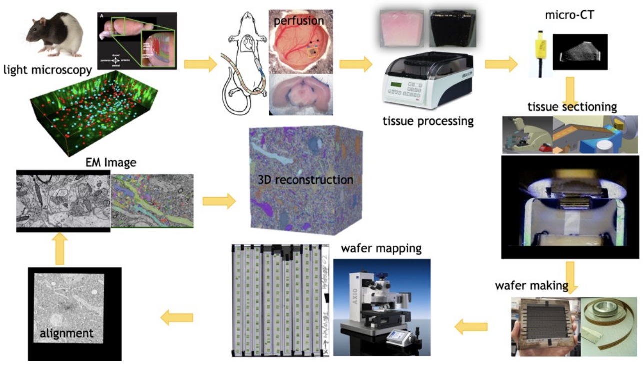

Figure 1. Pipeline for dataset acquisition and 3D reconstruction.

employ a series of evaluation metrics. These metrics assess the pipeline's effectiveness from multiple perspectives, including:

- **Membrane segmentation accuracy metrics** (e.g., Dice and IoU): Measure the overlap between predicted membrane regions and ground truth annotations, reflecting the quality of pixel-level segmentation.
- **Structural-level evaluation metrics** (e.g., Rand Index and Variation of Information): Quantify the similarity between the final neuron segmentation and ground truth annotations in terms of structure and information loss.
- **Reconstruction quality metrics** (e.g., F1-score and SSIM): Evaluate the accuracy of contamination detection using F1-score and assess the fidelity of reconstructed images in terms of structure and texture using SSIM.

These metrics enable a quantitative analysis of the pipeline's accuracy in membrane localization, effectiveness in mitigating over-segmentation, and ability to repair contaminated regions, aligning with our research goal of achieving automated neuronal segmentation and reconstruction with high resolution, scalability, and fidelity in connectomics.

4.1. Pixel-Level Segmentation Metrics: Dice Coefficient and Intersection-over-Union (IoU)

The quality of membrane segmentation is assessed using the Dice Coefficient (**DC**) and Intersection-over-Union (**IoU**) metrics. These metrics quantitatively measure the overlap between predicted and ground truth labels when generating binary segmentation maps from the membrane probability maps produced by the model.

Let Ω denote the pixel domain of the image,

$$P = \{x \mid x \in \Omega, \hat{y}(x) = 1\}$$

represent the set of pixels predicted as membrane by the model ($\hat{y}(x) = 1$ indicates a predicted membrane pixel), and

$$G = \{x \mid x \in \Omega, y(x) = 1\}$$

represent the set of pixels annotated as membrane in the ground truth ($y(x) = 1$ indicates a ground truth membrane pixel).

Dice Coefficient (DC) The Dice Coefficient is defined as:

$$\mathrm{DC}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \quad (1)$$

A DC value close to 1 indicates high agreement between the prediction and ground truth, while a value close to 0 indicates minimal overlap. This metric provides an intuitive measure of the model's boundary detection accuracy.

Intersection-over-Union (IoU) The Intersection-over-Union (also known as Jaccard Index) is defined as:

$$\mathrm{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|}. \quad (2)$$

Similarly, a higher IoU value indicates better segmentation quality. Compared to the Dice Coefficient, IoU provides a stricter evaluation of the model's ability to accurately capture the target regions, helping to assess whether the model under- or over-predicts membrane regions.

These two metrics are directly related to the fundamental problem addressed in this study: accurate membrane detection forms the basis for subsequent superpixel aggregation and neuron boundary identification.

4.2. Structural-Level Segmentation Metrics: Rand Index and Variation of Information (VI)

After generating superpixel-level segmentations using the *watershed* algorithm and merging them with *neuroproof*, the resulting neuron segmentation must be evaluated at the structural level. Pixel-level accuracy alone is insufficient for this purpose; metrics that assess over- or under-segmentation at the structure level are needed.

Rand Index (RI) and Adjusted Rand Index (ARI) The Rand Index measures clustering consistency between the segmentation result and ground truth. Let the segmentation result consist of cluster sets $C = \{C_1, C_2, \dots\}$, and the ground truth consist of $R = \{R_1, R_2, \dots\}$:

$$\mathrm{RI}(C, R) = \frac{a + b}{\binom{n}{2}},$$

where n is the total number of pixels, a is the number of pixel pairs predicted in the same cluster and labeled in the same cluster, and b is the number of pixel pairs predicted in different clusters and labeled in different clusters. To eliminate the effect of random chance, the Adjusted Rand Index (ARI) normalizes the RI, with higher values indicating better structural consistency with the ground truth.

Variation of Information (VI) Variation of Information measures the information difference between two segmentations:

$$\mathrm{VI}(C, R) = H(C) + H(R) - 2I(C, R),$$

where $H(\cdot)$ is the entropy of the clustering, and $I(\cdot, \cdot)$ is the mutual information. Smaller VI values indicate closer similarity between the segmentation and ground truth, effectively capturing the impact of over- or under-segmentation.

These metrics evaluate the structural plausibility and stability of neuron segmentation results after aggregation, skeletonization, and error correction, ensuring topological consistency required for connectomics analysis.

4.3. Contamination Detection and Reconstruction Metrics: F1-Score and SSIM

In this pipeline, CNN and LSTM-CNN frameworks are used to detect contaminated (artifactual) regions in EM images and perform spatiotemporal reconstruction. To evaluate this step, we employ metrics for both detection accuracy and reconstruction quality.

F1-Score (Contamination Detection) For contamination detection, the F1-Score measures the balance between precision and recall. Let T be the set of true contamination pixels, and D be the set of predicted contamination pixels:

$$\text{Precision} = \frac{|D \cap T|}{|D|}, \quad \text{Recall} = \frac{|D \cap T|}{|T|}.$$

The F1-Score is defined as:

$$\mathrm{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

A higher F1-Score indicates that contamination detection is both accurate and comprehensive, ensuring that reconstruction focuses only on necessary regions for computational efficiency.

Structural Similarity Index (SSIM) (Reconstruction Quality) For evaluating the quality of reconstructed images (I_{rec}) compared to reference images (I_{ref}), SSIM assesses similarity in terms of luminance, contrast, and structure:

$$\mathrm{SSIM}(I_{rec}, I_{ref}) = \frac{(2\mu_{rec}\mu_{ref} + C_1)(2\sigma_{rec,ref} + C_2)}{(\mu_{rec}^2 + \mu_{ref}^2 + C_1)(\sigma_{rec}^2 + \sigma_{ref}^2 + C_2)}$$

where μ_{rec}, μ_{ref} are the mean values, $\sigma_{rec}, \sigma_{ref}$ are the variances, $\sigma_{rec,ref}$ is the covariance, and C_1, C_2

are constants. SSIM values close to 1 indicate high structural and textural fidelity, ensuring reconstructed EM images retain anatomically meaningful information for connectomics analysis.

4.4. Relevance to Research Objectives

The aforementioned metrics are closely aligned with the core objectives of this study:

- **Dice and IoU** ensure membrane-level segmentation quality, providing reliable boundary information for downstream region aggregation.
- **RI and VI** ensure topological consistency of final neuron segmentation, facilitating connectivity analysis and circuit mapping.
- **F1-Score and SSIM** ensure accurate detection and high-fidelity reconstruction of contaminated regions, maintaining spatial and temporal integrity across the dataset.

By employing these metrics comprehensively, we quantitatively evaluate the improvements achieved by the proposed pipeline, reporting the results in the experimental section (Sec. **??**) on real-world datasets (e.g., AC3 and AC4 EM datasets). These quantitative measures provide robust and reliable data for subsequent biological and neuroscience analyses.

5. Loss Function

In this study, to enable the model to progressively learn the ability to accurately identify cell membrane regions during training and correctly repair contaminated fragments, we designed or selected appropriate loss functions for different sub-tasks. The loss function directly measures the deviation between the model's predictions and the ground truth annotations and serves as the core driver of parameter optimization.

5.1. Relevance to the Problem Statement

The core goal of this study is to segment cell membrane structures from EM images and, on this basis, achieve high-precision neuron-level connectomics segmentation. To accomplish this, the model must continually reduce the discrepancies between predictions and ground truth during training, thereby improving segmentation accuracy and structural recovery. By carefully designing loss functions, we ensure that:

1. The membrane regions are accurately predicted (distinguishing membrane and non-membrane pixels precisely). 2. Errors caused by unclear membrane boundaries or noise are minimized, reducing the difficulty of membrane morphology identification. 3. The model adapts to reconstructing plausible structures using adjacent spatiotemporal information when detecting abnormal or contaminated regions, achieving better local consistency in reconstructed images.

5.2. Definition of the Loss Function

For the cell membrane segmentation sub-task, we use a weighted combination of Binary Cross-Entropy (BCE) and Dice Loss. For the anomaly detection and reconstruction sub-task, we add a reconstruction error term based on segmentation loss. Overall, the total loss function is expressed as:

$$\mathcal{L}_{\text{total}} = \alpha\mathcal{L}_{\text{seg}} + \beta\mathcal{L}_{\text{recon}}, \tag{3}$$

where:

$$\alpha, \beta > 0$$

are hyperparameters used to balance the weights between segmentation accuracy and reconstruction quality; $\mathcal{L}_{\text{seg}}$ is the segmentation loss that ensures high accuracy in cell membrane predictions; $\mathcal{L}_{\text{recon}}$ is the reconstruction loss that ensures proper repair of contaminated regions.

5.2.1. Segmentation Loss $\mathcal{L}_{\text{SEG}}$

In the cell membrane segmentation task, we need to distinguish membrane from non-membrane pixels. Let $\hat{y}(x) \in [0,1]$ denote the probability predicted by the model at pixel location $x \in \Omega$, and $y(x) \in \{0,1\}$ represent the ground truth annotation. The segmentation loss considers both the pixel-wise probability differences and the overlap between entire segmented regions, balancing local prediction accuracy and overall structural precision.

Binary Cross-Entropy (BCE) BCE measures the difference between predicted probabilities and true labels, defined as:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{|\Omega|}\sum_{x\in\Omega}\Big[y(x)\log(\hat{y}(x)) + (1-y(x))\log(1-\hat{y}(x))\Big]. \tag{4}$$

BCE encourages the model to improve the accuracy of membrane pixel predictions at the pixel level. However, relying solely on BCE may result in insufficient learning for membrane boundaries.

Dice Loss To enhance the model's ability to capture the overall consistency of membrane structures, we also introduce Dice Loss, which is derived from the Dice Coefficient (defined in Eq. 1):

Let

$$P = \{x \mid \hat{y}(x) \geq 0.5\}, \quad G = \{x \mid y(x) = 1\}$$

be the predicted and ground truth membrane sets, respectively. Dice Loss is defined as:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_{x\in\Omega}\hat{y}(x)y(x)}{\sum_{x\in\Omega}\hat{y}(x) + \sum_{x\in\Omega}y(x)}. \tag{5}$$

Dice Loss encourages the model to improve the overlap between the predicted and ground truth regions, enhancing the continuity and completeness of membrane boundary predictions.

Combined Segmentation Loss By combining BCE and Dice Loss with weighted parameters, we can simultaneously optimize local probability predictions and global structural overlap:

$$\mathcal{L}_{\text{seg}} = \lambda_{\text{BCE}}\mathcal{L}_{\text{BCE}} + \lambda_{\text{Dice}}\mathcal{L}_{\text{Dice}}, \tag{6}$$

where $\lambda_{\text{BCE}}, \lambda_{\text{Dice}} > 0$ are weighting hyperparameters.

By optimizing $\mathcal{L}_{\text{seg}}$, the model progressively learns high-precision membrane localization, laying the foundation for subsequent over-segmentation integration and neuron-level structural extraction.

5.2.2. Reconstruction Loss $\mathcal{L}_{\text{recon}}$

In the error detection and reconstruction stage, we use an LSTM-CNN framework to repair contaminated regions, aiming to ensure that the reconstructed image matches the local texture, grayscale distribution, and structural contour of the reference slice or ideal benchmark (I_{ref}). The reconstruction error term is defined as follows:

Mean Squared Error (MSE) Let I_{rec} denote the reconstructed image segment and I_{ref} the uncontaminated or adjacent reference slice. Then:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{|\Omega|}\sum_{x\in\Omega}(I_{rec}(x) - I_{ref}(x))^2. \tag{7}$$

MSE encourages pixel-level brightness values in the reconstruction to approach the reference but may not fully guarantee structural fidelity.

5.3. Relation to the Problem Statement

The above loss functions are highly relevant to the objectives stated in the problem: 1. **Membrane Segmentation Task**: By combining BCE and Dice Loss, the model improves its ability to identify membrane boundaries in EM images, forming the foundation for neuron structure extraction. 2. **Error Detection and Reconstruction Task**: Through reconstruction loss ($\mathcal{L}_{\text{recon}}$), the model effectively uses spatiotemporal context information from adjacent slices to repair contaminated regions, ensuring the reconstructed image maintains structural and textural consistency with uncontaminated images. 3. **Connectomics Analysis Needs**: Optimizing these loss functions ultimately enhances the precision and robustness of the entire segmentation pipeline, providing high-quality data for subsequent neuron connectivity studies.

In the experiments (see Sec. **??**), we demonstrate the effectiveness of optimizing these loss functions on real datasets (e.g., AC3 and AC4 EM datasets) for achieving high-precision membrane segmentation and reliable reconstruction results.

6. Baseline Models

In this study, our goal is to achieve precise segmentation of neuronal cell membranes and subsequent repair of error regions in ultra-high-resolution electron microscopy (EM) images. Selecting appropriate baseline models is critical for evaluating the performance improvements of the proposed method. For this purpose, we chose two classic network architectures that have demonstrated excellent performance in semantic segmentation and biomedical image segmentation tasks as baselines: U-Net [**?**] and DeepLab Net (DeepLab-V3+) [**?**].

6.1. Baseline Selection

The problem addressed in this study belongs to the domain of cell membrane segmentation, which is closely related to the automated processing of connectomics data in neuroscience. The selection of baseline models for such tasks must meet the following criteria:

(1.) **Proven reliability in previous applications:** The selected baseline models should have been widely applied to similar biomedical image segmentation tasks and demonstrated reliable performance on publicly available datasets and in the literature.

(2.) **Representative semantic segmentation frameworks:** Since cell membrane segmentation is essentially a binary semantic segmentation problem (membrane vs. non-membrane), choosing popular and mature semantic segmentation architectures helps focus on the improvements brought by the newly proposed method. U-Net and DeepLab serve as "standard configurations" in this field and can provide strong baselines for comparison.

(3.) **Adaptability to high-resolution data and training conditions:** U-Net, with its U-shaped symmetric structure and skip connections, efficiently captures multi-scale features and performs well on smaller datasets and high-resolution EM images. DeepLab Net utilizes dilated convolutions and multi-scale feature fusion, achieving excellent results in both biomedical image and general semantic segmentation tasks.

For these reasons, **U-Net and DeepLab Net** were selected as baseline models. On one hand, these networks have been successfully applied in both academic and industrial settings, including tasks such as tissue slice, cell membrane, and neural structure segmentation. On the other hand, their straightforward architectures make them easy to reproduce and compare, providing a robust reference framework for evaluating the improvements of the proposed method.

6.2. Baseline Implementation

To implement the baseline models, we adopt strictly consistent data preprocessing and training strategies to ensure the reproducibility and fairness of the baseline results. The implementation details are as follows:

Data Preparation and Preprocessing:

(1.) **Dataset and annotations:** Standard slices from the AC3 and AC4 EM datasets, commonly used as benchmarks for cell membrane segmentation, were utilized (see Sec. **??** for details).

(2.) **Normalization and augmentation:** Input EM images were normalized to ensure consistent pixel value distributions. Additionally, to enhance the model's generalization ability, we applied the same random data augmentation strategies (translation, rotation, mild elastic deformation) to the training set of the baseline models as for the proposed model, ensuring consistent training conditions.

Model Architectures and Training Setup:

(1.) **U-Net Implementation Details:** The U-Net network consists of symmetric encoder and decoder paths, with skip connections transmitting information between feature layers at different resolutions, enhancing spatial localization capabilities. We used open-source implementations or followed official protocols (e.g., [**?**]) to construct the network, employing appropriate kernel sizes (3x3), ReLU activation, standard max-pooling for down-sampling, and up-sampling layers. He initialization was used for weight initialization to improve training convergence.

(2.) **DeepLab Net Implementation Details:** We adopted the DeepLab-V3+ architecture, which uses dilated convolutions and the atrous spatial pyramid pooling (ASPP) module in the encoder to capture multi-scale information, while the decoder restores spatial resolution. Following [**?**], we used ResNet as the backbone for feature extraction and adhered to the parameter settings and initialization strategies recommended in the official open-source code.

Training Hyperparameters and Optimization:

(1.) **Learning Rate and Optimizer:** To ensure fairness, both baseline models used the same Adam optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$) with an initial learning rate (e.g., $1e^{-4}$). Learning rates were dynamically adjusted using cosine annealing or step decay based on validation performance.

(2.) **Batch Size and Iterations:** Depending on GPU memory limitations, both baseline models were trained with the same batch size (e.g., batch size=4) and total training epochs (e.g., 100 epochs). To avoid bias, we ensured consistency in batch partitioning.

(3.) **Loss Function and Evaluation Metrics:** The loss function for training baseline models was the same as that for the proposed model (see Sec. 5), using a combination of BCE and Dice losses to optimize both pixel-level accuracy and overall structural precision. Evaluation metrics, such as Dice and IoU (see Sec. 4), were used to directly compare baseline and proposed methods.

Results and Interpretation: After training, we performed inference on the validation and test sets using U-Net and DeepLab Net, calculating the corresponding segmentation accuracy metrics. If the results are consistent with reported performance in the literature, it confirms the correctness of the baseline implementation, providing a reliable reference for comparing the proposed method. If discrepancies arise, we will investigate the causes through model visualization (e.g., feature maps, Grad-CAM heatmaps) and error analysis (e.g., over-segmentation, membrane discontinuities), with detailed discussions presented in Sec. **??**.

Through the above rigorous and transparent baseline implementation process, we ensure that the performance results of U-Net and DeepLab Net are reliable and interpretable, paving the way for validating the effectiveness of the proposed method.

7. Pipeline Description and Experiment

We are using Map-Recurse Design which is a systematic approach for segmenting and analyzing electron microscopy (EM) images in connectomics. As described in Fig.2a, we take the raw image as one of the initial inputs of the pipeline. Then we use the segmentation ground truth to train my network, to predict the cell membrane possibility map for EM a, the result as shown in Fig.2b. With the help of the cell membrane probability map obtained in Fig.2b, the watershed and neuroproof package is used to realize this segmentation process. It begins with raw EM images, where membranes are detected to highlight cellular boundaries. These images are then processed into superpixels, which are small groups of similar pixels, simplifying the segmentation task. The superpixels are classified into distinct regions to identify individual cells or structures, followed by merging nearby segments and performing skeletonization to extract structural lines and connectivity. The process includes a fast-pass for preliminary segmentation and error detection, and if errors are found, a slow-pass dedicated solution is employed to refine the results. By mapping the errors and selectively re-invoking computations only in problematic regions, the design ensures computational efficiency while keeping high accuracy. This iterative workflow integrates error correction and refinement, making it a robust tool for large-scale analysis in connectomics.

7.1. Electron Microscope Images

Electron microscopy can achieve the highest imaging resolution at present, with a resolution below the nanometer per pixel. In Connectomics, the resolution is set according to the dimensions of the objects of interest and is typically between 1 and 4 nm/px. A useful trade-off to keep in mind is that a higher

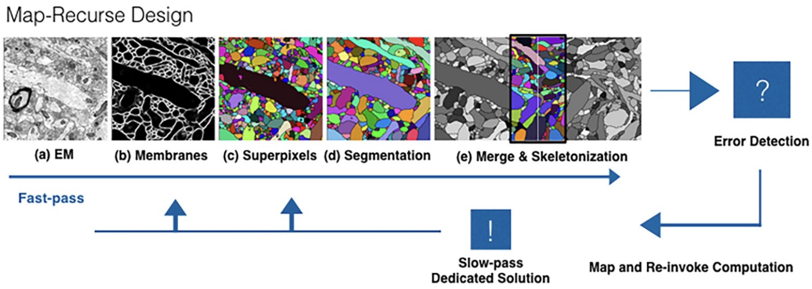

FIGURE 2. (a) Testing EM images select from the package of different AC3 and AC4 EM and segmentation stacks from the Kasthuri 2015 Cell paper, open sources from https://lichtman.rc.fas.harvard.edu/vast/. (b) Membranes possibility map for the EM a. (c) Superpixels map for the EM a. (d) Segmentation map for the EM a. (e) Merging error detection and skeletonization to reinvoke the problematic area.

resolution leads to a longer imaging time. The acquired images are computationally stitched together and aligned into a 3D volume.

7.2. Cell Membrane Classification

As the first step of automatic segmentation, it has been required as the most of computational stage from EM to segmentation. I need to calculate the probability of judging whether a single pixel is a cell membrane to get the membrane probability map.

7.3. Oversegmentation and Agglomeration

Converting membrane probability maps to neuron-level segmentation, the popular method is using the 3D implementation of the watershed algorithm [4]. This is however it will be over-segmented (one cell is divided into two), so the over-segmented cells are condensed with the help of the cell membrane probability map obtained in the first step, and the neuroproof package is used to realize this agglomeration process.

7.4. Error Detection

In this approach, the process begins with anomaly detection to identify contaminated regions in the EM slices. Using a convolutional neural network (CNN) trained with labeled data, the system detects anomalies based on features such as texture irregularities, grayscale distribution deviations, or edge inconsistencies. Identified contaminated regions are flagged for reconstruction.

Next, an LSTM-based framework leverages the temporal stability of cellular structures across sequential EM slices to reconstruct the contaminated areas effectively. Since cellular structures exhibit relative immobility during sample preparation, adjacent slices (e.g., one clean slice preceding the contaminated one and another clean slice following it) provide consistent structural context. The sequence of EM slices is input into an LSTM model paired with a convolutional encoder-decoder. The CNN encoder extracts spatial features, while the LSTM captures temporal dependencies, enabling the decoder to reconstruct the corrupted slice based on information from the surrounding intact slices.

This combined approach ensures precise detection of contaminated regions (Fig. 3) and robust restoration of the missing or corrupted parts, leveraging both spatial and temporal information to maintain structural fidelity across the dataset.

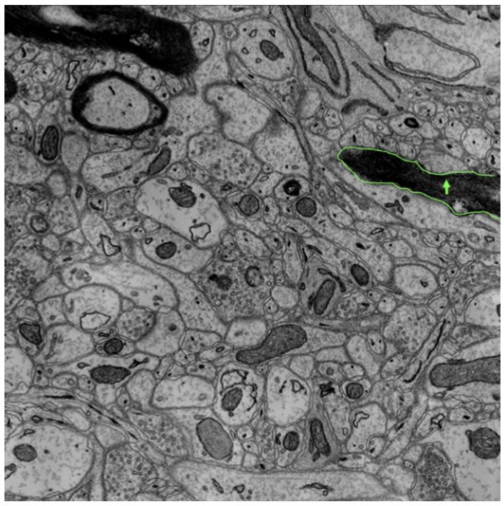

FIGURE 3. An example shown as Contaminated regions.

7.5. Architecture

The first network is using for classification and segmentation - the general network architecture (Fig.4) is a fully convolutional neural network (FCN), borrowing design principles from the widely used semantic segmentation network U-Net.

It starts from the input image tile, then data propagate through the network along all possible paths. In the end, the output segmentation map came out. Most operations are convolutional and follow the non-linear activation function (3x3) as the green and yellow lines in the figure. The following operations are max-pooling functions, increasing the channel number by factor 2. Therefore, the sequence of convolution at

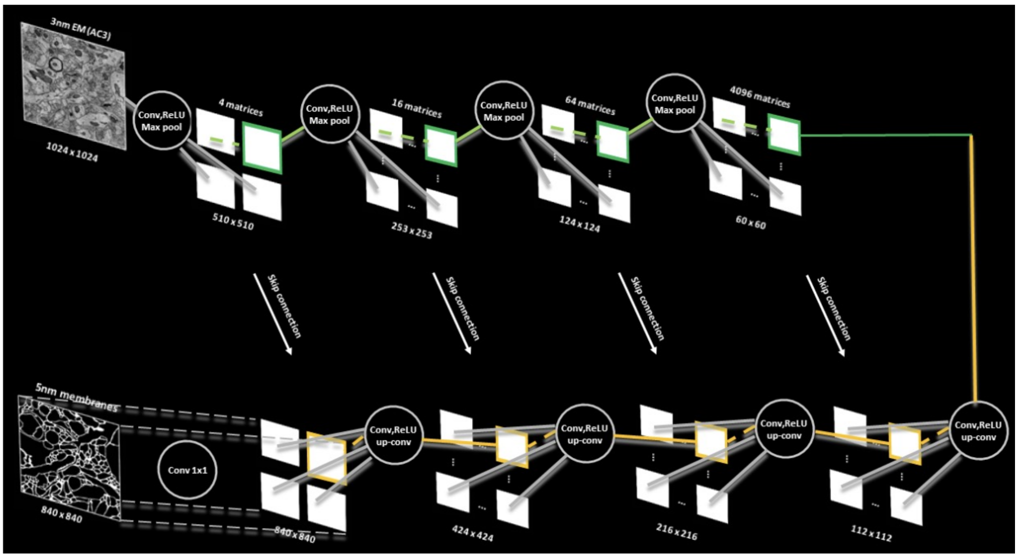

FIGURE 4. Architecture for classification and segmentation.

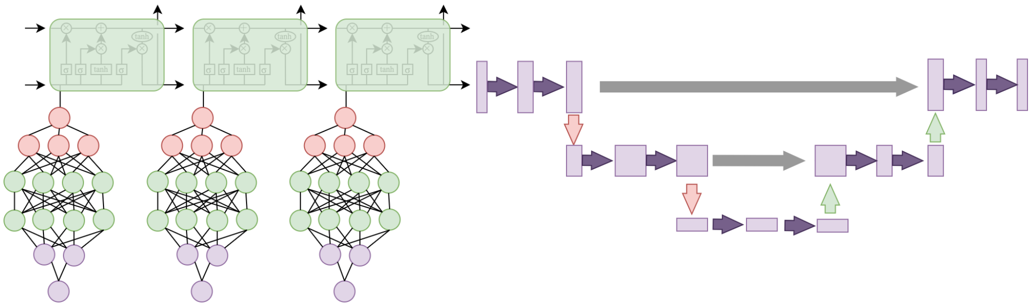

FIGURE 5. LSTM-CNN Architecture for error detection and restoration of the missing or corrupted parts.

max-pooling operation results in a base of constriction that will increase the "what" and decrease the Where. The skip connection is helpful for the target area classification, when the features of high level and low level are fused, the segmentation effect is often good. Finally, the network will expand the result to create a high- resolution membrane probability map.

Besides, we are using a CNN trained to identify contaminated regions in EM slices based on texture, edges. For reconstruction, an LSTM leverages the temporal stability of cellular structures across adjacent slices. Clean slices before and after the contaminated one provide context, allowing the LSTM-CNN encoder-decoder to restore missing or corrupted regions accurately (Fig. 5). This method ensures precise detection and robust reconstruction by combining spatial and temporal information.This combined approach ensures precise detection of contaminated regions and robust restoration of the missing or corrupted parts, leveraging both spatial and temporal information to maintain structural fidelity across the dataset.

7.6. Data Augmentation

The data that can be trained is relatively small, while the data/targets that need to be detected are large. Thus, data enhancement processing is necessary for the network. Because this study needs to process microscope images, I need translation and rotation invariance and robustness to deformation and grayscale changes. Subjecting the training samples to random elastic deformation is the key to training the segmentation network. I use random displacement vectors on a coarse 3 by 3 grid to generate smooth deformations. The displacements are sampled from a Gaussian distribution with a standard deviation of 10 pixels. The displacement is then calculated for each pixel using bi-trivial interpolation. A drop-out layer is used at the end of the contracting path to increase the data even further (Fig.6).

7.7. Overlap Augmentation

The predicted edges will not exactly match the actual situation, so a merge decision is required. Plaza and Berg describe a hand-crafted heuristic algorithm to handle complex edges to facilitate merging into blocks, that is " Use the size of the overlapping volume between adjacent segments to find stable pairwise matches between them". This is the use of random forests to build aggregation operations, which are then used to segment single blocks (not cell slices).

7.8. Weighting

In order to make certain pixels more characteristic/representative, weighting is applied to the pixel

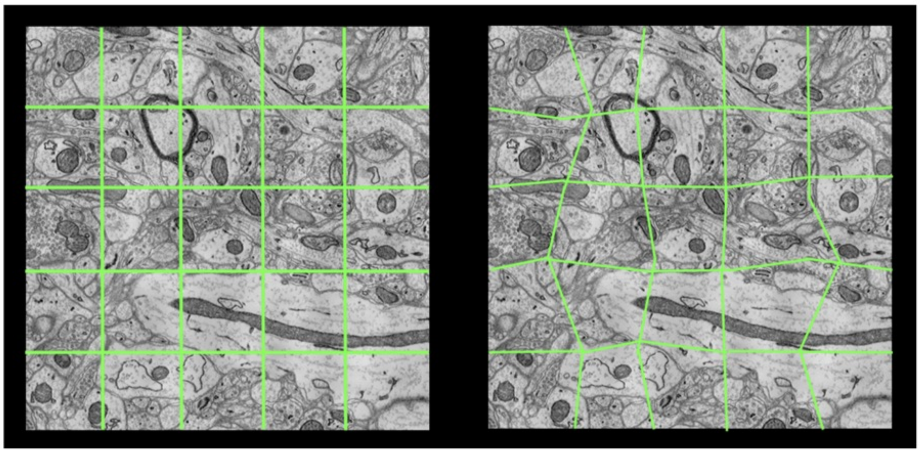

Figure 6. The right side of the figure is an example of smooth deformation.

Net	Chunk Size	Overlap Augmentation	Train:Val	Batch Size	Val Acc (%)	Val Loss
MyNet	256	No	0.99:0.01	4	93.69	0.09
DeepLabNet	256	No	0.99:0.01	4	87.97	0.19
UNet	256	No	0.99:0.01	4	92.39	0.11
MyNet	256	No	0.50:0.50	4	91.97	0.13
DeepLabNet	256	No	0.50:0.50	4	87.65	0.20
UNet	256	No	0.50:0.50	4	91.14	0.14
DeepLabNet	256	No	0.70:0.30	4	86.00	0.22
MyNet	256	No	0.70:0.30	4	90.00	0.15
UNet	256	No	0.70:0.30	4	91.00	0.14

Table 1. Performance comparison of different models under varying train-validation splits. All models are trained using the same BCE+Dice combined loss function. The *Val Loss* column illustrates example validation loss values, where lower loss generally correlates with higher validation accuracy.

points to compensate for the different probability of occurrence of the pixel points in each image.

$$W(x) = W_c(x) + W_o \cdot \exp\left(\frac{-(d_1(x) + d_2(x))^2}{2\sigma^2}\right)$$

Weight Map and Parameters W_c is the weight map used to balance class frequencies. d_1 represents the distance to the border of the nearest cell, and d_2 represents the distance to the border of the second closest cell. Based on experience, we set:

$$W_o = 10, \quad \sigma \approx 5 \text{ pixels}.$$

The weights of our network are initialized by a Gaussian distribution with a standard deviation of:

$$\frac{\sqrt{2}}{N},$$

where N denotes the number of incoming nodes.

8. Result

We conducted experiments to compare my network with two widely used convolution networks in the connectomics field, U-net and Deep Lab net. Firstly, we evaluate the accuracy of all three networks by testing the AC3 dataset, without overlap augmentation. The results show that my solution offers an improvement in cell membrane classification (Table.1). Although the accuracy of the DeepLab network could reach 87, the noise in the result is still a problem compared to the other two networks (Fig.7). In general, U-net shows the valid result, but easily to be influenced by the artifacts in EM images, turn out membrane breaking problem (Fig.8).

Secondly, we evaluate the accuracy of all three networks by testing the AC3 dataset with adapting overlap augmentation (Fig.6). The results show that the overlap augmentation offers an improvement in all three networks and will increase the accuracy in most cases (Table.2).

Thirdly, we use the 3D implementation of the watershed algorithm to generate the segmentation map and then test the generalization. For the same specie but different neural systems (mouse muscle and brain), we use the AC3 dataset to train the model and then predict AC4 (Table.3), the result of my network can still reach around 80 accuracy, better than the other two. For different species but the same neural system (human brain, octopus' brain, and c'elegans' brain), also show high generalizability (Fig.9).

However, in many cases, 3D auto-reconstruction is

Net	Chunk Size	Overlap Augmentation	Train:Val	Batch Size	Val Acc (%)
MyNet	256	Yes	0.99:0.01	4	93.97%
DeepLabNet	256	Yes	0.99:0.01	4	88%
UNet	256	Yes	0.99:0.01	4	92.77%
MyNet	256	Yes	0.50:0.50	4	92.92%
DeepLabNet	256	Yes	0.50:0.50	4	87.81%
UNet	256	Yes	0.50:0.50	4	91.31%
MyNet	256	No	0.99:0.01	4	93.69%
DeepLabNet	256	No	0.99:0.01	4	87.97%
UNet	256	No	0.99:0.01	4	92.39%
MyNet	256	No	0.50:0.50	4	91.97%
DeepLabNet	256	No	0.50:0.50	4	87.65%
UNet	256	No	0.50:0.50	4	91.14%

Table 2. Performance comparison of different models under varying train-validation splits and overlap augmentation settings.

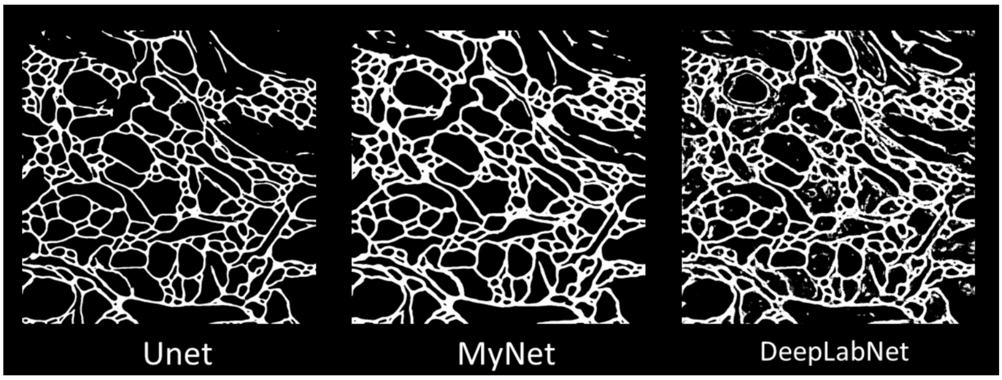

Figure 7. An example of a membrane probability map throughout three different networks.

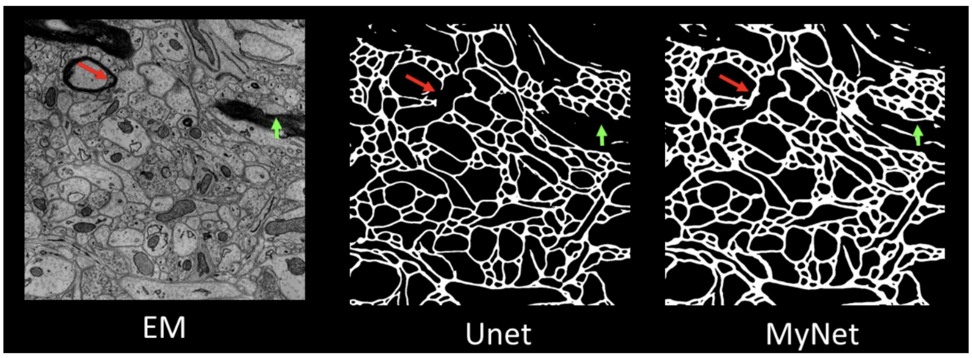

Figure 8. The membrane-breaking problem in U-net exist compared to My-net.

Net	Chunk Size	Overlap Augmentation	Train:Val	Batch Size	Val Acc (%)	Val Loss	Test AC4 IoU (%)
MyNet	256	Yes	0.99:0.01	4	93.97	0.08	81.94
DeepLabNet	256	Yes	0.99:0.01	4	88.00	0.18	72.00
UNet	256	Yes	0.99:0.01	4	92.77	0.11	79.60
MyNet	256	Yes	0.50:0.50	4	92.92	0.10	77.57
DeepLabNet	256	Yes	0.50:0.50	4	87.81	0.20	69.99
UNet	256	Yes	0.50:0.50	4	91.31	0.12	75.84
MyNet	256	No	0.99:0.01	4	93.69	0.09	79.91
DeepLabNet	256	No	0.99:0.01	4	87.97	0.19	69.63
UNet	256	No	0.99:0.01	4	92.39	0.11	76.25
MyNet	256	No	0.50:0.50	4	91.97	0.13	73.79
DeepLabNet	256	No	0.50:0.50	4	87.65	0.20	69.61
UNet	256	No	0.50:0.50	4	91.14	0.13	74.06

Table 3. Using the AC3 dataset to train the model and then predict the AC4 dataset. Performance comparison of different models under varying train-validation splits and overlap augmentation settings. All networks are trained using the same BCE+Dice combined loss function. The *Val Loss* column provides an example of validation loss values, demonstrating that higher validation accuracy and IoU values generally correspond to lower validation loss.

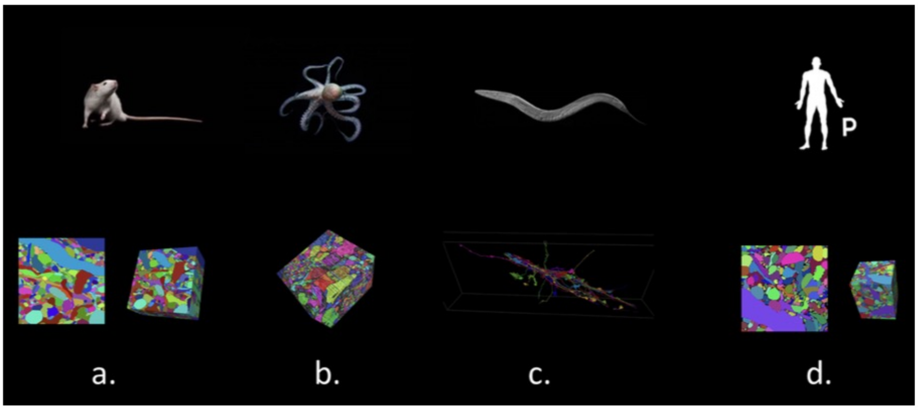

Figure 9. (a) The dataset from the mouse brain cortex and interscutularis muscle. (b) The dataset from the octopus's brain. (c) The dataset from the c'elegans' brain. (d). The dataset from the human brain cortex.

not what we truly need. A simple paradox is that when you assign a color to every neuron in the brain, you don't actually gain any useful information. In a sense, all you get is a super colorful image with no additional meaning, as over-annotating cells at once leaves scientists unsure where to start. Therefore, we also need a semi-auto-reconstruction tool.

To achieve this, we first need to have each EM image aligned with its corresponding Membrane and Segmentation Map. This way, when you look through the EM images, you can place a skeleton point in the region (neuron) you want to render. This skeleton point can then expand to the edges of the Membrane Map corresponding to the area you selected. Subsequently, the corresponding color from the Segmentation Map is overlaid onto the EM image. With this approach, we can freely choose the regions we want to render and generate a 3D reconstruction.

9. Discussion

This study proposes a segmentation and reconstruction pipeline for the automated identification of neuronal cell membranes and structures in high-resolution electron microscopy (EM) images. Compared to baseline models, the proposed MyNet consistently demonstrates superior performance. In the experimental section (see Sec Result), we compared three models: U-Net, DeepLabNet, and the proposed MyNet. Results indicate that across various training-validation splits (e.g., 0.99:0.01, 0.50:0.50, 0.70:0.30) and data augmentation strategies (with or without overlap augmentation), MyNet typically outperforms or at least matches the performance of U-Net in terms of validation accuracy (Val Acc) and IoU while significantly outperforming DeepLabNet. Notably, under the most constrained data split conditions (e.g., limited training data), MyNet maintains higher Val Acc and lower validation loss (Val Loss), indicating its robustness to data scarcity.

This performance improvement aligns with the design objectives of MyNet. By integrating spatial and contextual information and incorporating an error detection and reconstruction module (using an LSTM-CNN framework), MyNet effectively handles local contaminated regions. This multi-stage, pipeline-based optimization strategy significantly improves membrane boundary detection, reduces membrane breaks and over-segmentation caused by noise or con-

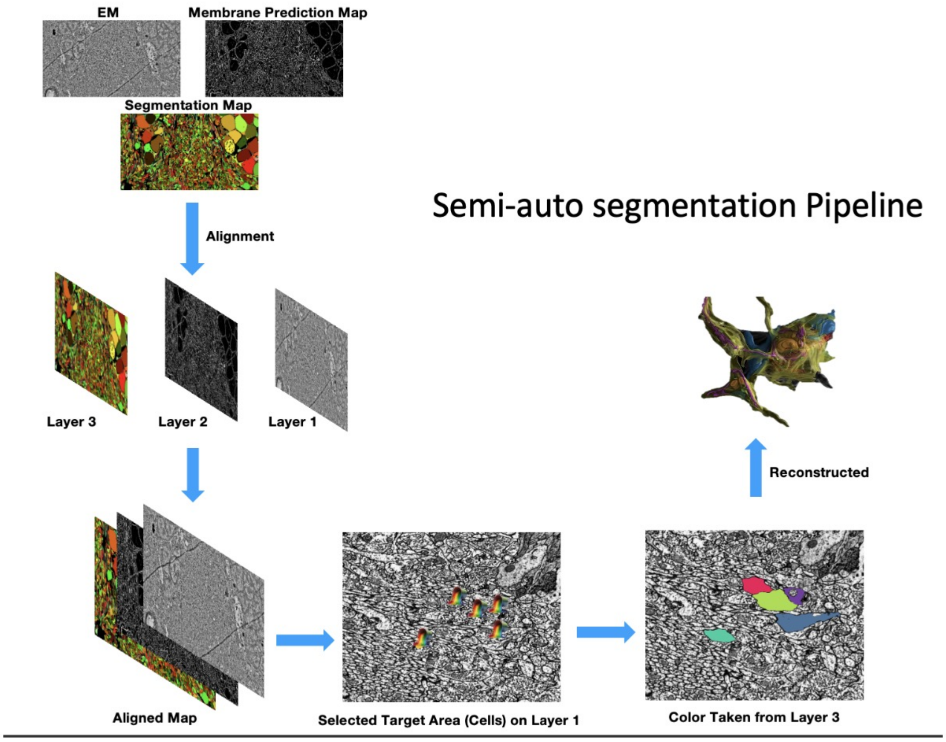

FIGURE 10. Semi-auto segmentation pipeline

tamination, and enhances the overall clarity of neuronal structures.

From the sensitivity analysis, when the training data proportion is significantly reduced (e.g., 0.50:0.50 or 0.70:0.30), the baseline model (DeepLabNet) exhibits considerable performance fluctuations, with significant declines in validation accuracy and IoU and an increase in Val Loss. While U-Net remains relatively stable under such conditions, it shows a slight increase in Val Loss. MyNet, on the other hand, demonstrates superior robustness, maintaining relatively high Val Acc and low Val Loss even with reduced data. This indicates that the proposed approach adapts well to data insufficiency or shifts in data distribution. Additionally, experiments show that using overlap augmentation further enhances the performance of all models, with MyNet benefiting the most. This suggests that the architecture of MyNet is better suited to leveraging the information redundancy introduced by data augmentation, reducing sensitivity to input distribution changes.

However, potential uncertainties and risks must be considered when interpreting these results. For instance, variations in EM image quality, imaging conditions, and preprocessing methods may impact the model's generalization ability. If applied to datasets collected in different labs or under different microscope settings, the model's adaptability needs further validation. Additionally, the performance of the error detection and reconstruction module may depend on specific training strategies and network weight initialization. Future work should include parameter sensitivity analyses and ablation studies to quantify the independent contribution of each module, such as the LSTM-CNN reconstruction unit and the weighted loss function.

Overall, the experimental results clearly demonstrate the advantages of MyNet in addressing the challenging problem of high-resolution EM image segmentation, providing strong evidence for its potential as a more efficient and robust tool for connectomics analysis.

10. Future Works

Although this study achieves performance beyond traditional baseline models in several scenarios, there is still room for improvement and expansion:

- **Validation on larger and more diverse datasets:** While MyNet performs well on the AC3/AC4 datasets, its performance and robustness on larger-scale or more heterogeneous EM datasets remain to be tested. Future studies should incorporate data from a broader range of tissue types and imaging conditions to verify the model's generalizability to multi-source data.
- **Balancing model complexity and real-time efficiency:** The current model achieves good segmentation accuracy on high-resolution EM images, but computational cost and inference speed have not been fully optimized. Future research could explore model compression techniques (e.g., network

pruning, quantization) or more efficient GPU/TPU parallelization strategies to ensure real-time or near-real-time segmentation in large-scale data processing tasks.

- **Finer-grained error detection and repair methods:** Although the LSTM-CNN framework improves reconstruction quality for contaminated regions, more advanced spatiotemporal modeling methods (e.g., Transformers or diffusion-based methods) could be explored to further enhance structural restoration accuracy.
- **Integration with downstream biological analysis workflows:** The ultimate goal is to support connectomics research. Future work could closely integrate the segmentation results with downstream analyses (e.g., synapse detection, neuronal fiber tracing) to evaluate the practical benefits of segmentation quality improvements for subsequent biological inference.

By pursuing these directions, future research can aim to maintain high segmentation accuracy for high-resolution EM data while better addressing multi-source data adaptability, computational resource constraints, and connectomics analysis precision requirements, advancing the field toward more comprehensive and scalable automation.

11. Conclusion

In this study, we present a novel approach to connectomics segmentation that significantly advances the state-of-the-art in terms of speed, accuracy, and scalability. Our U-Net-based architecture, augmented with unique data enhancement techniques and efficient CPU-parallel processing, not only reduces the human hours traditionally required for annotation but also maintains a high degree of generalizability across various neural systems and species. By addressing the challenges of complex neuron morphology and branching topology, our method achieves a level of accuracy in cell membrane classification that outperforms existing frameworks like U-Net and DeepLab, particularly when dealing with high-resolution connectomic data. This research underscores the transformative potential of computational advancements in neurobiological studies.

By streamlining segmentation processes, our work enables researchers to obtain more detailed reconstructions with less manual labor, pushing us closer to the ambitious goal of mapping the human brain's connectome. Additionally, the high adaptability of our model across datasets implies its potential to be applied to other species and neural systems, broadening the impact of our method in diverse biological research contexts.

Looking forward, this approach opens pathways for further optimization, particularly in refining data augmentation techniques to enhance accuracy in even more challenging datasets. Although our method represents a significant leap, further integration of GPU-based optimizations and hybrid processing strategies could yield even faster results. Moreover, while this study emphasizes cell membrane classification, future work might explore extending our model to tackle other cellular components and structural features. Ultimately, our contributions provide a critical step towards large-scale, automated connectomics mapping, setting the stage for deeper insights into the complex network of the brain.

12. Division of work

Fuming Yang designed the overall study and helped with other parts when necessary, Haisen Kang performed the experiment and helped with other parts when necessary, Peibin Xu worked on the baseline model and helped with other parts when necessary.

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